

Machine Learning and Earth Observation for the Resilience to Weather Extremes in the Mediterranean

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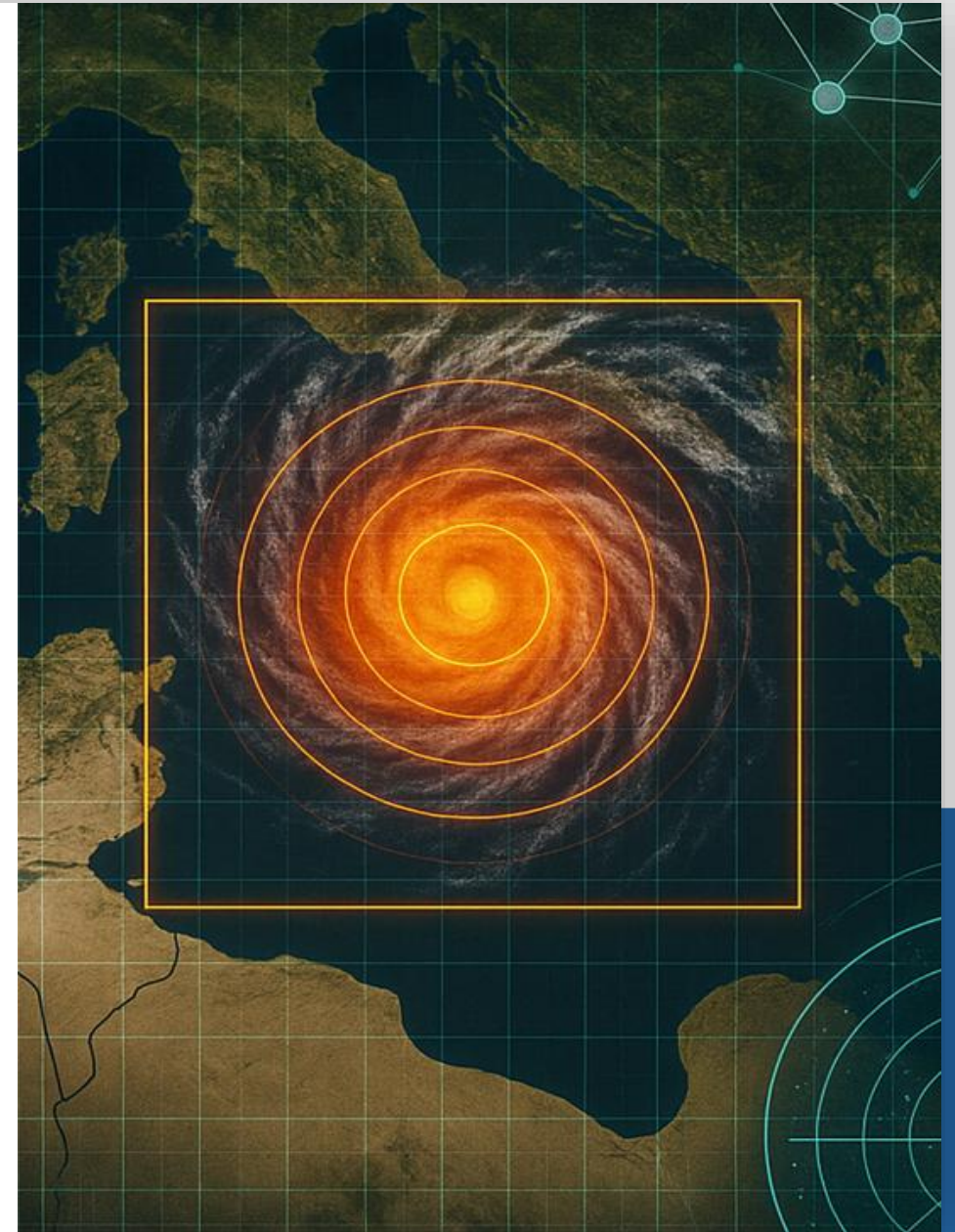
Mentored by Marcello Miglietta, Giulia Panegrossi,
Leo Pio D'Adderio, Stefano Sebastianelli, Paolo Sanò, Daniele
Casella, Andrea Camplani, and Daniele D'Armiento (CNR-ISAC)



High Meadows
Environmental
Institute



Consiglio Nazionale
delle Ricerche



The Institute of Atmospheric Sciences and Climate (CNR-ISAC)

- CNR-ISAC branch located in Rome, Italy 🇮🇹
- Institute of the *Italian National Research Council* (CNR)
- Aim to enhance our understanding of the *atmosphere*, *ocean*, and *climate* through a *multidisciplinary approach*
- We collaborated within the Medicanes Project, an initiative funded by the *European Space Agency (ESA)* for leveraging Earth observations to understand and predict extreme weather (i.e. tropical-like cyclones) in the Mediterranean



CNR-ISAC (Rome-based) satellite team

Medicanes and their Impact

- Medicane: Mesoscale cyclones developing in the Mediterranean region that display characteristics similar to Tropical Cyclones (TCs)
- 1-3/year, relevant to high impact weather, producing severe damages due to strong wind gusts, high sea waves and heavy precipitation

Example: Medicane Daniel (September 2023)

- Affected cities in Greece and Libya and excessive rainfall collapsed two dams
- Over 5,000 fatalities and 10,000 people missing. Buildings, communication lines and power networks fell



*Storm Daniel devastation in
Greece & Libya
(Sep. 5-10, 2023)*

*Images by Vaggelis Kuosioras
@ Associated Press*



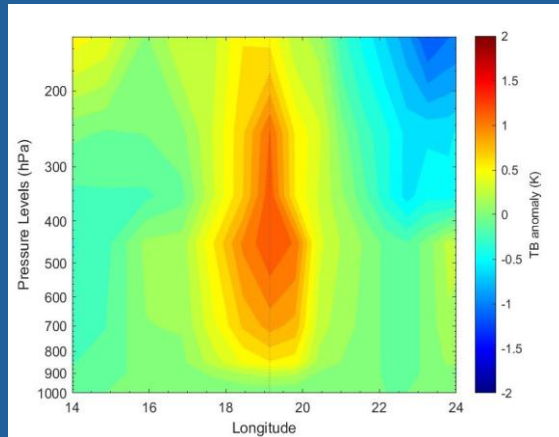
Project Goals

Miglietta et al., BAMS, 2025

- Develop automated tools for identification and monitoring of medicanes using Earth Observation (EO)
- Improve assessments of environmental risks and socio-economic impacts associated with Medicanes

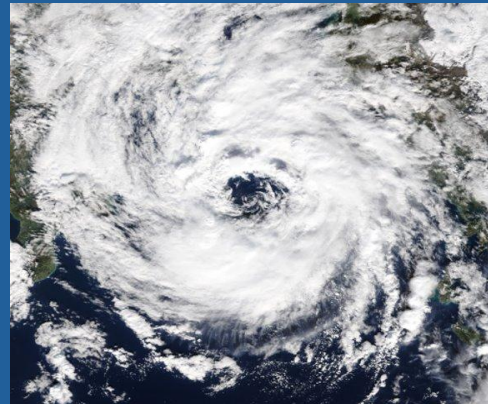
Defining characteristics of Medicanes:

Warm core (WC)



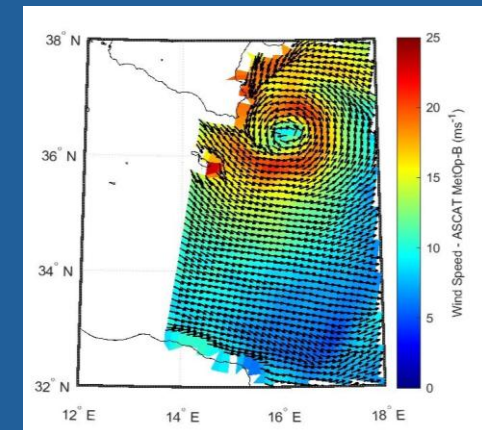
MW obs. of TB anomaly - Medicane Zorbas
(29 Oct. 2018)

Almost-cloudless "eye" with spiraling cloud structure and rain bands



MODIS Terra VIS image - Medicane Numa
(28 Nov. 2017)

Nearly closed-ring surface wind field



ASCAT wind field - Medicane Apollo
(29 Oct. 2021)

Deep Learning for Automatic WC Detection

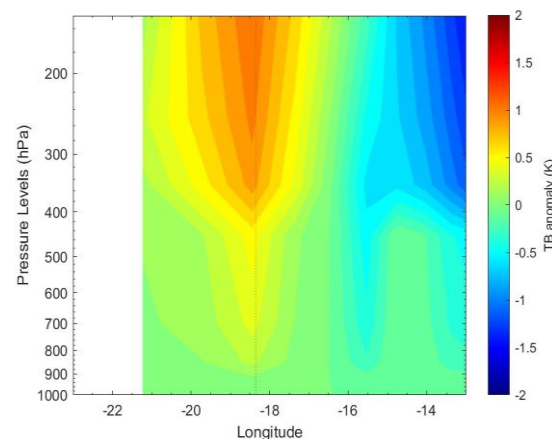
Panegrossi et al., Remote Sens., 2023
Plots created by Leo Pio D'Adderio (CNR-ISAC)



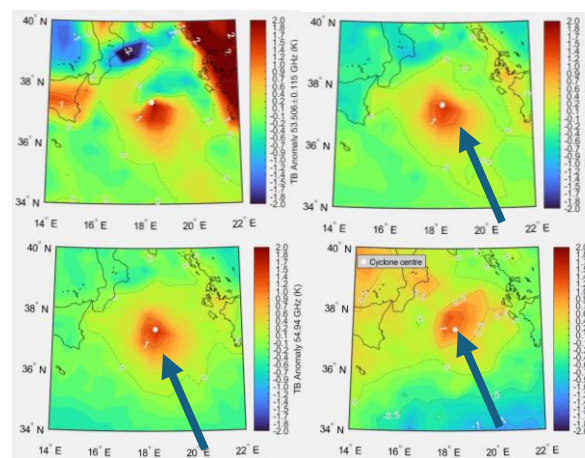
- WC: region of warm temperature extending from the middle to the upper troposphere at the center of the storm
- Traditionally recognized using satellite microwave radiometers and visual inspection for presence of nearly isolated, positive Brightness Temperature (TB) anomaly across temperature sounding channels



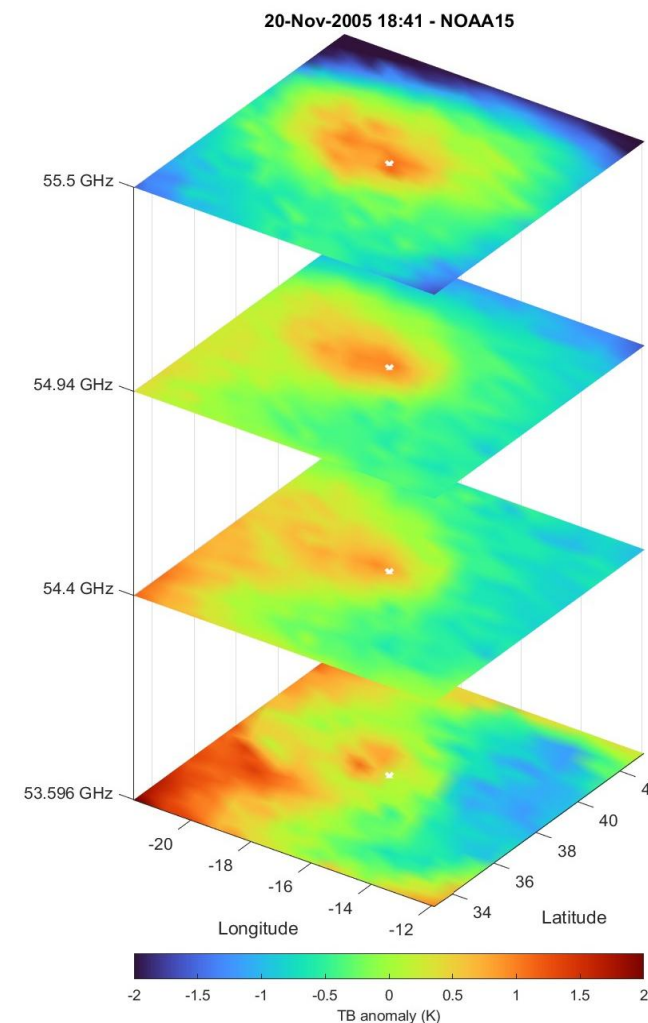
- Predict presence of key Mediane characteristic in near real time with deep learning analysis
- Improve weather warnings for informed community decision making



Vertical cross section
(cross-track)

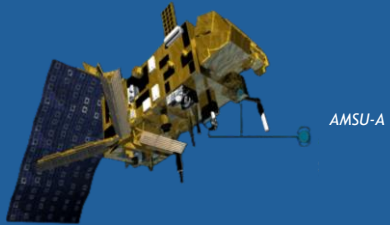


Extremely Deep WC (2D plots)



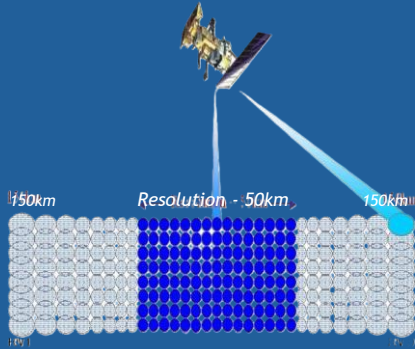
3D plot

Preprocessing Radiometer Data

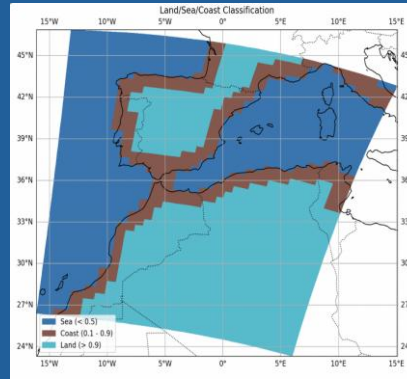


AMSU-A

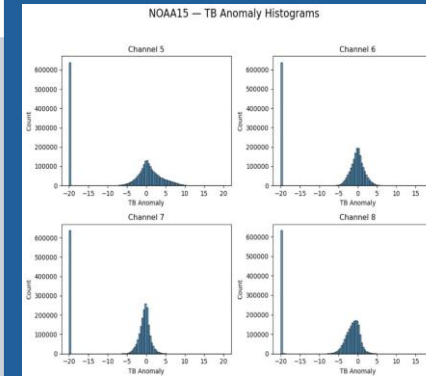
Passive Microwave Radiometry data (from instruments AMSU-A, ATMS, SSMIS)



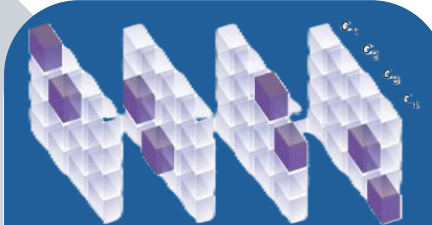
Cyclone center near edges of swath (low resolution)



Cyclone center proximity to land and coast



Corrupted instrumentation

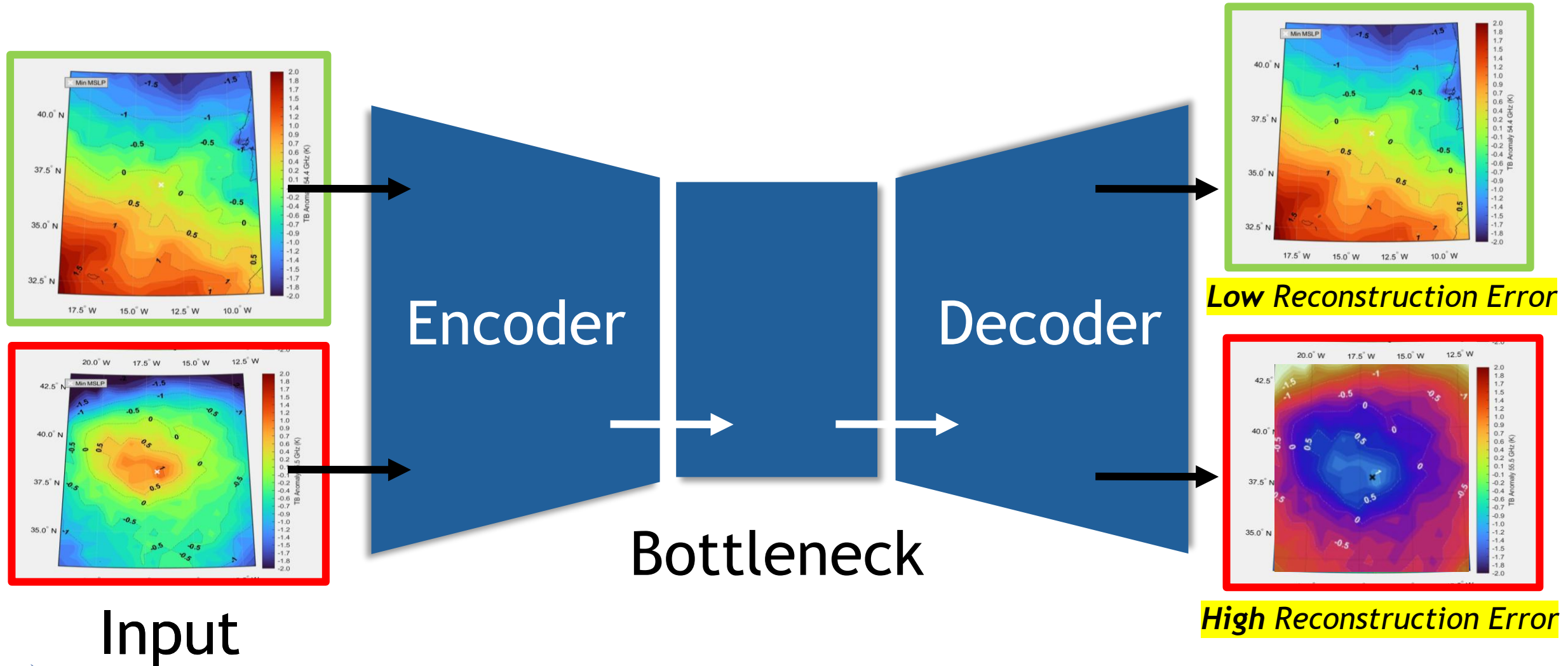


Incorporate metadata and prepare tensor

Criteria for filtering out overpasses (data) due to instrument limitations

Deep Learning Approach: AE-SAD

Angiulli et al., pre-print, 2023



Results

Dataset

450+ cyclones with 12K overpasses across 3 instruments

~25% of overpasses from raw data are cleared for use in training and testing

~15% warm core (positive/anomaly)

Architecture

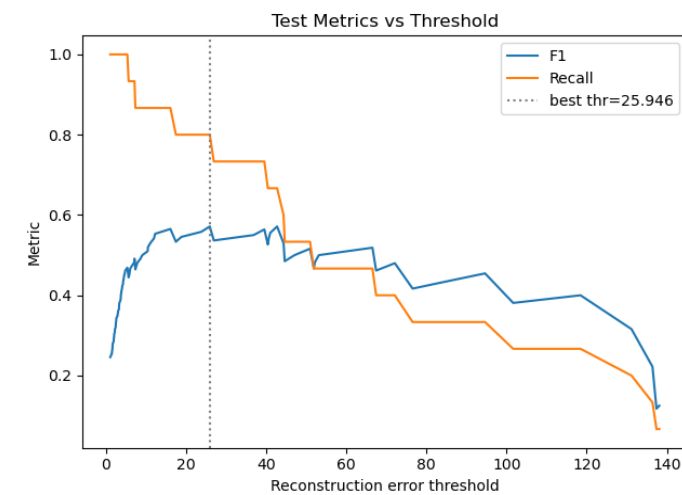
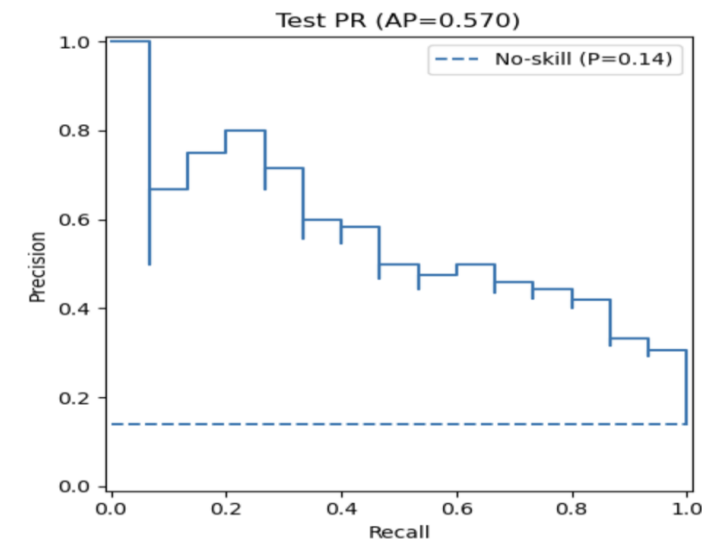
- Convolutional autoencoder

Parameters:

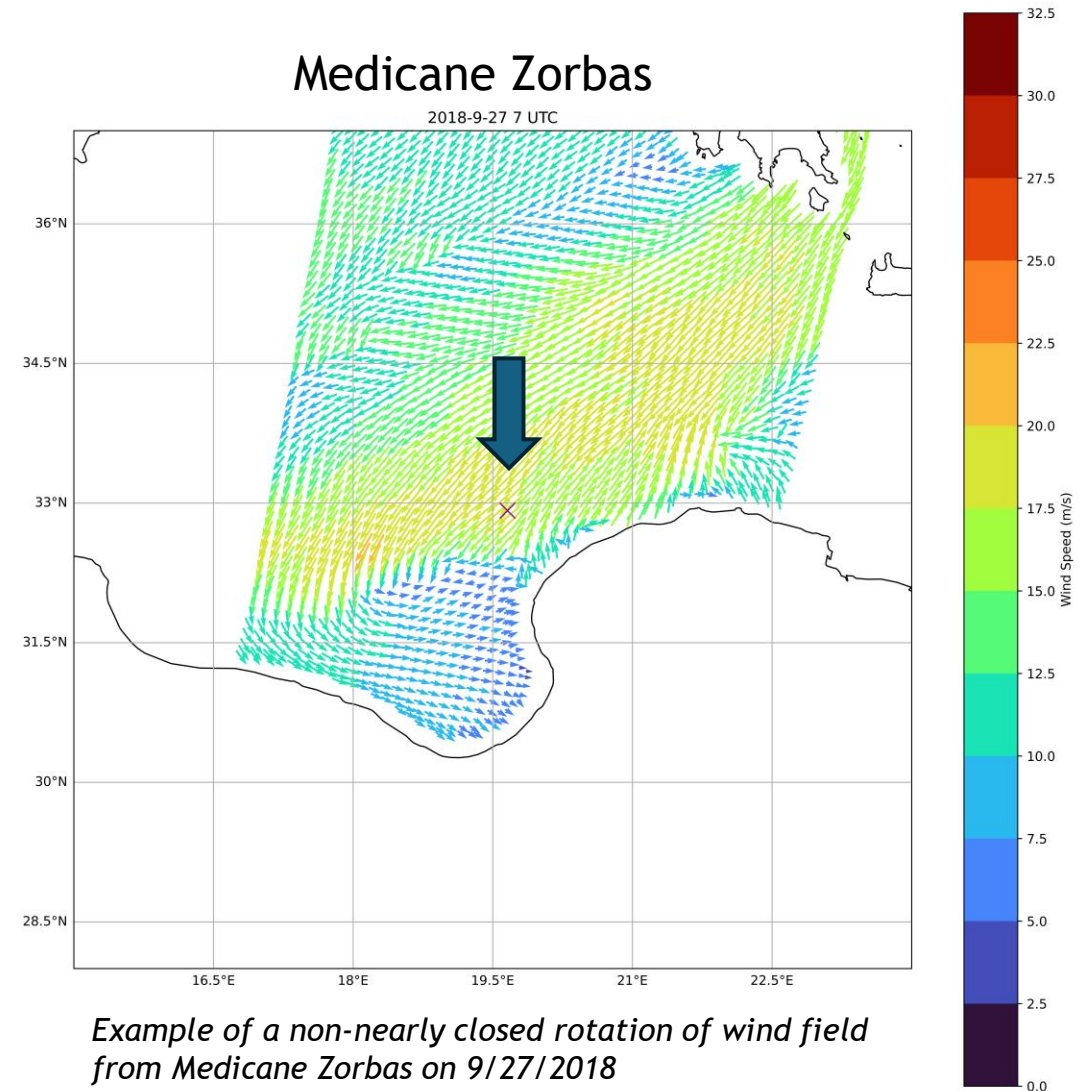
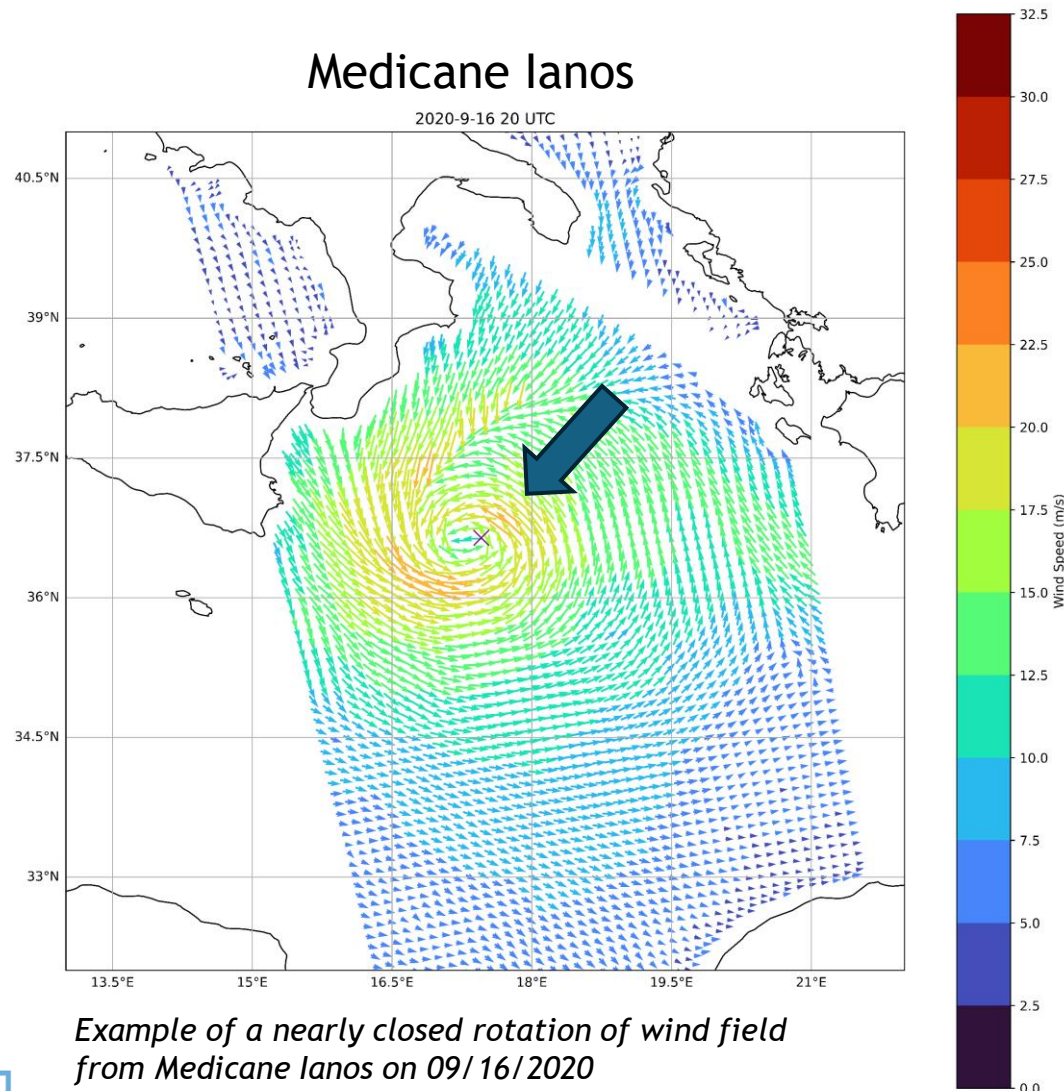
- Weight assigned to reconstruction of anomalies, latent dimension

Performance

- **AUC** (Area Under Curve) of **Precision-Recall curve** for binary classification task with highly imbalanced dataset
- Compared to the baseline positive class fraction of **0.14**, the model achieved an AUC-PR of **0.789**, representing a substantial improvement



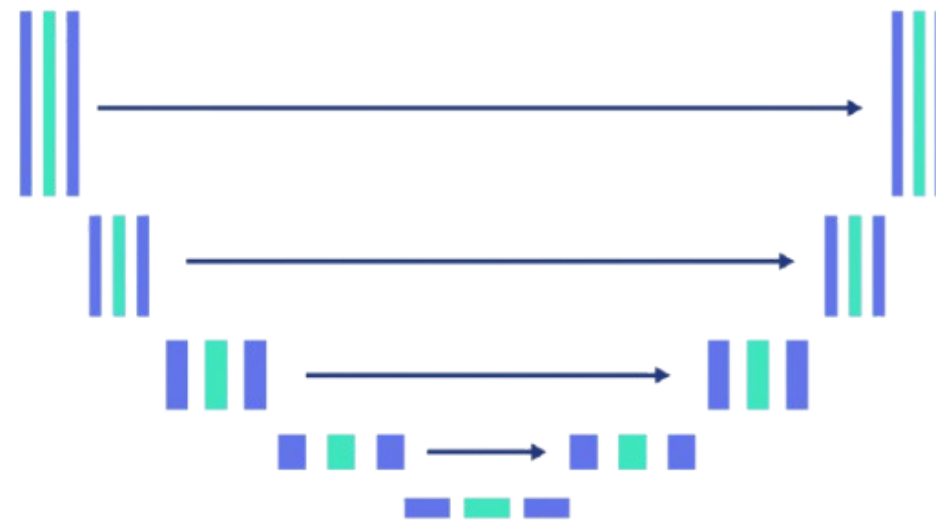
Nearly Closed Surface Wind Field of Medicanes



Machine Learning Pipeline: U-Net

Ronneberger et al., MICCAI, 2015

- **Model:** U-Net
- **High-level concept:** shrinks an image to learn its features, then scales it up its original dimensions to make predictions, while using skip connections to preserve finer details
- Replaced convolution layers with partial convolution layers ([Liu et al., ECCV, 2018](#)) to account for missing wind data outside the swath and over land



Architecture of U-Net

Preprocessing ASCAT Data

Data from European MetOP satellites

- 161 x 161 matrix of wind direction and wind speed is collected along the two swaths. No wind data is recorded over land, between the swaths, or outside the swaths
- Preprocessed to 80 x 80 matrix of horizontal (U) and vertical (V) components of wind vectors. 791 overpasses from 2010-2020.
- Points that are 100 km from the cyclone center serves as the **segmentation mask**

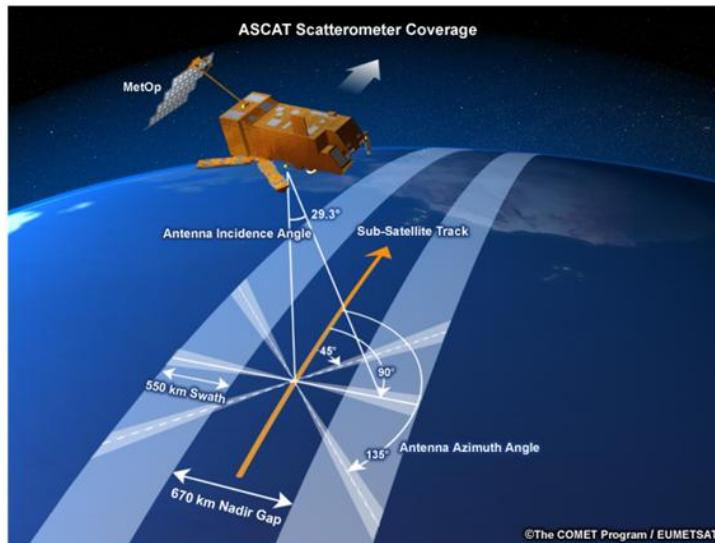
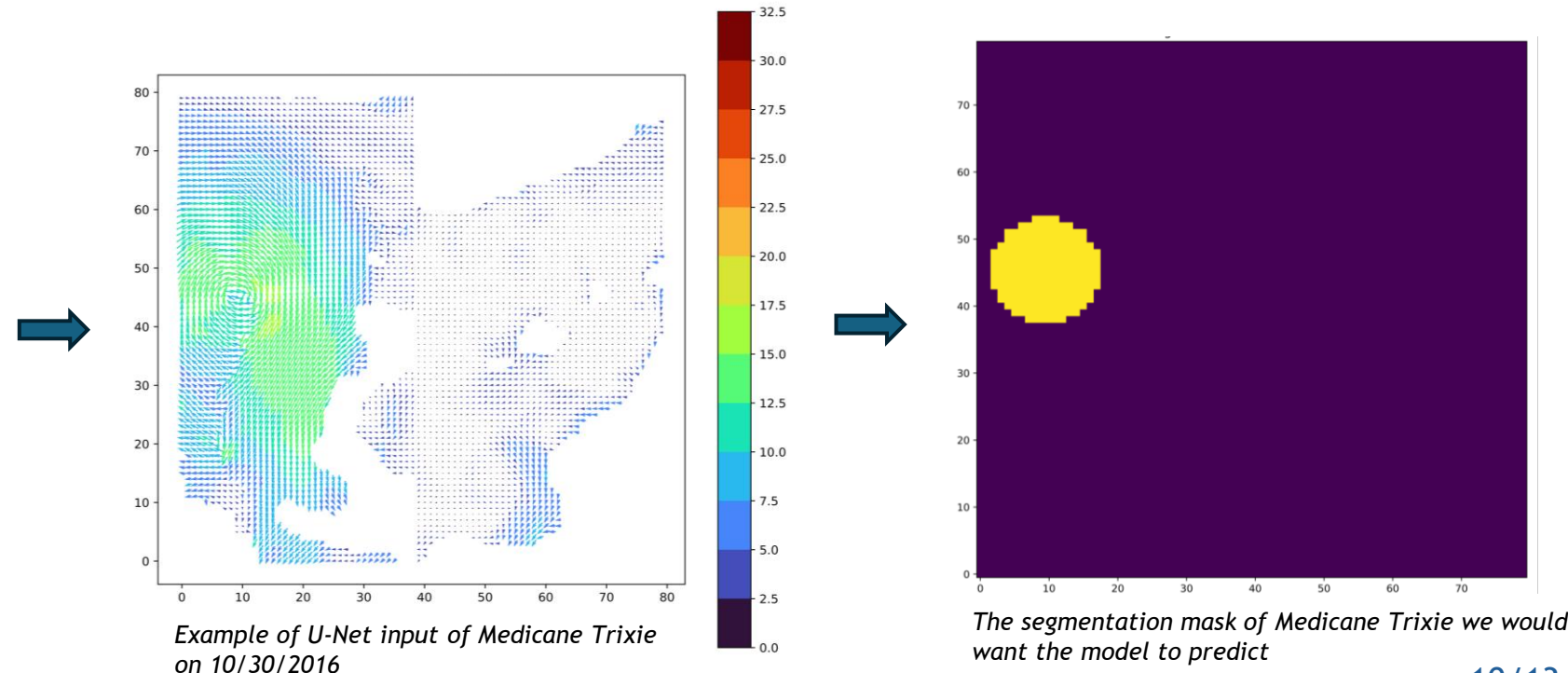


Illustration on ASCAT coverage of the Earth



Example of U-Net input of Medicane Trixie on 10/30/2016

The segmentation mask of Medicane Trixie we would want the model to predict

Results

Precision	Recall	mIoU of background	mIoU of labeled
0.4981	0.5000	0.9963	0.0000

Interpretation

- Because the dataset is **heavily imbalance**, the U-Net model tends to guess a pixel is part of the background instead of part of the closed-ring of wind field

Total number of background pixels	Total number of labeled pixels
5,044,965	17,435
~99.7%	~0.3%

Next Steps

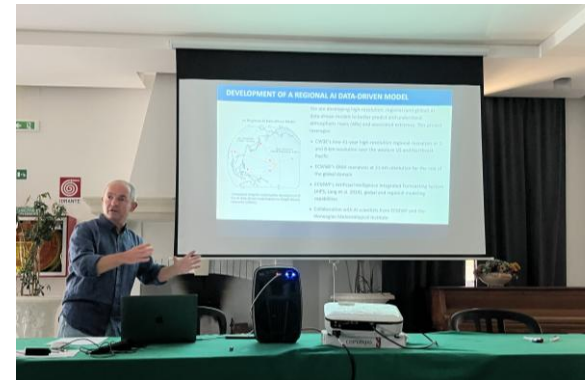
- Use **classification models** instead of semantic segmentation models, like AE-SAD
- Look into **imputation methods** to deal with missing data instead of using partial convolution layers

CNR-ISAC Conference

- Theme: *Severe Weather in a Changing Climate*
- Attended 20+ sessions discussing AI, remote sensing and data analysis, and systems-engineering approaches to resilience
- Networked with 35+ researchers, professors, and industry leaders from 10+ countries (e.g. Scripps Institution of Oceanography-UCSD, McGill University, University of Genova)



Conference Participants (June 30 - July 4, Castro Marina)



Prof. Delle Monache (UCSD) shares his pioneering work with AI models for monitoring atmospheric phenomena



Learning about research highlights from PhD presenters

Reflections & Future Steps

1 Personal Reflections

- Gained hands-on experience at the intersection of computer science and environmental studies, exploring how machine learning can advance climate and cyclone research
- Learned from researchers across career stages (from master's students to senior scientists) about academic research and real-world applications

2 Next Steps

- Aim to continue optimizing our model accuracy through data refinement and testing new algorithms
- Plan to share our results with the academic community through publication and integration with the Mediane project team's contributions

Acknowledgements

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<https://medicanes.isac.cnr.it/>

“Earth Observations as a cornerstone to the understanding and prediction of tropical-like cyclone risk in the Mediterranean”

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